# Exercise 2: SQL Aggregate Functions & Operators

~~Question 1~~ ~~Question 1~~

## Section 1: The Students Table

### Question 1

```
SELECT DISTINCT department
FROM students;
```

~~Department~~

| department |
| --- |
| IT |
| HR |
| Finance |

### Question 2

~~Sele~~

```
SELECT department,
       AVG(age) AS avg_age
GROUP BY
FROM students
GROUP BY department;
```

| department | avg_age |
| --- | --- |
| IT | 20,5 |
| HR | 22 |
| Finance | 23 |

Question 3:

```
SELECT department,
       COUNT(*) AS student_count
FROM students.
GROUP BY department
HAVING COUNT(*) > 1;
```

| department | student_count |
|---|---|
| IT | 2 |
| HR | 2 |

Question 4

```
SELECT student_id,
       name,
       age,
       department
~~WHERE~~ FROM students
WHERE age BETWEEN 21 AND 23;
```

| student_id | name | age | department |
|---|---|---|---|
| 2 | Bob | 22 | HR |
| 3 | Charlie | 21 | IT |
| 4 | Diana | 23 | Finance |
| 5 | Eve | 22 | HR |

Question 5

```
SELECT student_id,
       name,
       age,
       department
FROM students
~~WHE~~ WHERE department IN ('IT', 'HR')
~~WHE~~ AND age > 21
```

| student_id | name | age | department |
|---|---|---|---|
| 2 | Bob | 22 | HR |
| 5 | Eve | 22 | HR |

~~Question 6~~

Section 2 - The Courses Table.

Question 6

```
SELECT department,
       SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5 ;
```

| department | total_credits |
|---|---|
| IT | 11 |

Question 7.

```
SELECT course_id,
       course_name,
       department,
       credits
FROM courses
WHERE credits <> 4;
```

| course_id | course_name | department | credits |
|---|---|---|---|
| 101 | SQL Basics | IT | 3 |
| 104 | Excel | Finance | 2 |
| 105 | Statistics | HR | 3 |

Question 8

```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

| course_id | course_name | credits |
|---|---|---|
| 102 | Python | 4 |
| 103 | Data Science | 4 |
| 101 | SQL Basics | 3 |

## Secbion 3 - The enrollmenbs Table

### Quesbion 9

```
SELECT MAX(grade) AS max-grade,
       MIN(grade) AS min-grade,
       AVG(grade) AS avg-grade
FROM enrollmenbs;
```

| max-grade | min-grade | avg-grade |
|---|---|---|
| 90 | 78 | 84,60 |

### Quesbion 10

```
SELECT course_id,
       COUNT(enrollmenb_id) AS enrollment_count
FROM enrollmenbs,
GROUP BY course_id;
```

| course_id | enrollmenb_count |
|---|---|
| 101 | 1 |
| 102 | 1 |
| 103 | 1 |
| 104 | 1 |
| 105 | 1 |

Section 4 - The Salaries Table.

## Question 11

```
SELECT department,
       SUM(salary) AS total_salary,
       SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

| department | total_salary | total_bonus |
|---|---|---|
| IT | 122000 | 10500 |
| HR | 109000 | 7500 |
| Finance | 70000 | 6000 |

## Question 12

```
SELECT department,
       AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

| department | avg_salary |
|---|---|
| IT | 61000 |
| Finance | 70000 |

Question 13:

```
SELECT employee_id,
       name,
       salary,
       bonus,
       ~~(salary + b~~
       SUM(salary, bonus) AS total_compensation
FROM salaries
WHERE SUM(salary, bonus) > 60000
```

| employee_id | name | salary | bonus | total_compensation |
|---|---|---|---|---|
| 1 | Tom | 60000 | 5000 | 65000 |
| 3 | Spike | 70000 | 6000 | 76000 |
| 4 | Tyke | 62000 | 5500 | 67500 |

Section 5 - The projects Table

Question 14

```
SELECT department,
       SUM(budget) AS total_budget,
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;
```

| department | total_budget | avg_budget |
|---|---|---|
| IT | 270000 | 135000 |
| | | |

Question 15

```
SELECT project_id,
       project_name,
       department,
       budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';
```

| project_id | project_name | department | budget |
|---|---|---|---|
| 1 | AI App | IT | 120000 |
| 2 | Payroll System | Finance | 80000 |
| 5 | HR Portal | HR | 50000 |